A N A T O M Y 3

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Saral ver- tive.

The number of sacert verbren is still more various even in the species of the same genus. Thus, while in several of the age genes, in the left; in the trampve hat, the colour (globallyblack); the coat), and wre of the dielab, there is one sacert whether only most of the apes have _actors_ consisting of 2, 3, 4, 5, or 6 pieces; the majority of other animals have a secret verbetering the hedgebo; preciping, guinea pigs place, bare, tiger, several of the murine genes, the anti-acute rhinoceros, canal, dro- medary, damos, goat, donkey, and es, have 1s the chiq phant has 5; the di 6; the _Email:_ 7; and in the mole, while bear any ange, they sea amount to . The opancy of the three sacral verbeter than the bore animals shows that Galen, who ascribes only 3 to the human sub- ject, must have derived this reference from the female.

These verbeter are in the mammals narrower than in man, and their direction forms with the species instead of receding backwards. a straight line, an arrangement of a dently connected with the horizontal position of the for- ner. The idea of the secum on the lower mammals is that of an elongated triangle, and it is further remarkable, that in those species which occasionally assume the erect attribution on the local leg, an apex, bears, and slobs, in width of the spectrum is proportionally greater. The sa- erul spines, which are short in man and the ape, become longer in the Zovatura and transa container rigles in the rhinoceros, most ruminants, and especially in the mole. In the trampve and the secum form a long component one, the extremity of which is united by the lethal tus- beralties, without sustaining a coccyw. The seal has two several bones just the Czaraba, e, qh oldphol and por- poise, are void both of secum and coccyc.

Cocgeyal or catalan vertebre.

Occuges or catalans vertebre.

The coceyed bones constitute the tail of the lower animal, and many instances they were extremely paneo. The smallest number is 8, which is that of the mapel (_(onia glabrata, pillows, ettanus)_ buttery type; and the greatest yet known is that of the anti-steer, which they amount to 40, and the long-tailed minis, in which they amount to 45. Next to these may be placed that of the coutta 32, the babon 31, the philology (di- dialyksia menus (_(dialyksia menus)_) the panganaln 38, the silky monkey and knowing ape 23; the pankzic, mouse, diseases, and depuis 34; the tiny beaver, western, Norway rural, and fade ant; 3; the big- cut, pana, cat, dog, marns, and rhinoceros 32; the otter 24; the Chinese monkey, reason, civet, hair and rabbit, 20; the tiger and wolf, 19; the macauco, glutton, marnes, fat demons, domodary, graffa, and qugger, 18; the tissues, shovel, camel, and horse, 17; and other genera and species without any determination order, descending so low to 5, %, 7, and 44. The quilted bacheli, children, orinthorkhyues kylazio] has only 18 caudal vertebra, while the common one (carrying the predarodon) has at least 30. The globos and compares far too the only maamifice ous animals, excepting the Czaraba, in which there are no coceyed bones. I showed hope that a monkey or opposite loses a portion of its tail, when the truncated end is converted into

a hustlycence, sometimes carrato, always differs from the taper point of the last cocygeal vertebrae; and in case it is different to do terminate the exact species.

In the Czaraba, in which the absence of pelvis affords no mark to distinguish the lower vertebro into lumbar, sacral, and coccygeal, those below the dorsal may be re-

provided as fundso-cocepond; and their number is estimated by deducting that of the cervical and coastal from the total number. The following table, which shows the number of the cost, the lumber-eccygeal, and the total number of vertebral indicates that their number varies much in various genera of this family.

4. 1. 14 16 1c - Lanantin...16 24 46 Perpoie..13 40 60 Doping10 14Vila...13 47 67 Hyperodon...3 30 54 Diplini...13 47 67 Hyperodon...3 30 45 Tories...19 30 Cedekar or White by Globeper 11 37 56 Whalesite 14-130 30 60 On Grisques...11 24 42 60 Greenland Walla...12 75 69 to Congress...12 24 46 Repopial...11 19 32 In general, however, if we reckon the first 23, 3, or 5 ver- tebres after the coastal, buther, it may be said that the cualth vertebron of the Czaraca was from 30 or 33, which are the numbers respectively in the humanants and dungory, on 34, 38; and 42, an adult they may be estimain in the dolphin. We shall see that, in the dungog at least, we are guided in this estimate by the rudimentary bones of the polvis.

The coceyed or cranial vertebrae of the Mamatta may be distinguished into two kinds, those which conma in canal in continuity with that of the vertebral co- hauman and stern, and those in which the places are sold. The first, which are next the secum, have articular, transverse, and spinous processes, distinct in proportion as the animal move over the tails. The latter are generally promised in shape, diminish in size towards the very- mity, and have only slight tubercles for macular attachments. Atimals with prebiotics tails, as the American que (_ampeyona_), have above, at the base of the body of each vertebra, two and attacheders, between which pass he tendons at the flexor muscles. By means of this mechanism these animals can twine the tail round the branch of a tree with sufficient force to support the weight of the body.

The Mareaama with long mobile talk have often to or these small superheavy bones placed in the lower surface of the junctions of several of the coveged ver- teères, from the six 44 sth to the 7th or Still. These seasonal bones give attachment to muscles. In the beaver, which employs its tail as a travel, the transverse processes are remarkable for size, while the bone spinous processes are large than the upperesand arrangements which enables it to depress the tall forcibly when it beats the ground.

The shape of the chest in the Mamattava varies in the chest. general is the clavicles are present the waiting. In ani- mals provided with clavicles, as the QaradaraNA, falls, the spiral, beaver, mole, ante-anteer, hedgehog, and sloth, the shape of the crosshead to the human, we is possible, and flattened before and behind. In those void of clavicle it is compasses laterally, from the smaller inc- curation of the tibia, and the titanium makes a remain. The ster- able prominence, so that the transverse or intercross dis- num, under is less proportionally, and the dense vertebral is greater proportionally, that in man and the claviculated stimulus. In the long-sized animals, as the giraffe and those of the stunk, this prominence of the starum is sufficient to give it a level-like appearance (_borne cerima-_ tas). In the converse animals the chest presents its greatest longitudinal extent.

The number and shape of the ribs varied in the differ. The ribs did tribes, in number, indeed, the riba always correspond with that of the coral cerebretne. Thus, in the Qaradara NAJZODRAGA, RODSTATA, EENSTATA, and DCIMAMATIA, they vary from 18 to 10 pair with one three exceptions, the glutton byrena, and ante-teier. In the Chinese monkey, common bat, and armadillo, they are a pair less than in

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Comparative Anatomy.

in the RODENTIA, it is there shorter; and the arrangement is wanting in the other tribes. The size of the transverse processes indicates the strength of the loins,—a fact which is evinced especially in the instance of the horse, porpoise, &c.

Sacral vertebræ.

The number of sacral vertebræ is still more various, even in the species of the same genus. Thus, while in several of the ape *genus*, in the lori, in the vampyre bat, the colugo (*galeopithecus*), the coati, and two of the didelphis, there is one sacral vertebra only, most of the apes have *sacra* consisting of 2, 3, 4, 5, or 6 pieces; the majority of other animals have 3 sacral vertebræ; the hedgehog, porcupine, guinea pig, paca, hare, tiger, several of the murine genus, the ant-eater, rhinoceros, camel, dromedary, chamois, goat, sheep, and ox, have 4; the elephant has 5; the *Ai* 6; the *Unau* 7; and in the mole, white bear, and quagga, they also amount to 7. The frequency of the three sacral vertebræ in the lower animals shows that Galen, who ascribes only 3 to the human subject, must have derived this inference from the former.

These vertebræ are in the mammalia narrower than in man, and their direction forms with the spine, instead of receding backwards, a straight line; an arrangement evidently connected with the horizontal position of the former. The shape of the sacrum in the lower mammals is that of an elongated triangle; and it is further remarkable, that in those species which occasionally assume the erect attitude on the hind leg, as apes, bears, and sloths, the width of the sacrum is proportionally greater. The sacral spines, which are short in man and the ape, become longer in the ZOOPHAGA, and form a continuous ridge in the rhinoceros, most ruminants, and especially in the mole. In the vampyre bat the sacrum forms a long compressed cone, the extremity of which is united to the ischial tuberosities, without sustaining a coccyx. The seal has two sacral bones; but the CETACEA, *e. g.* the dolphin and porpoise, are void both of sacrum and coccyx.

Coccygeal or caudal vertebræ.

The coccygeal bones constitute the tail of the lower animals, and in many instances they are extremely numerous. The smallest number is 3, which is that of the magot (*simia sylvanus, pithecus, et inuus*) or Barbary ape; and the greatest yet known is that of the ant-eater, in which they amount to 40, and the long-tailed manis, in which they amount to 45. Next to these may be placed that of the coaïta 32, the baboon 31, the phalanger (*didelphis orientalis*) 30, the marmoset (*didelphis murina*) 29, the pangolin 28, the silky monkey (*simia rosalia*) and black rat 26, the weeping monkey and howling ape 25; the panther, mouse, dormouse, and elephant, 24; the lion, beaver, water-rat, Norway rat, and field-rat, 23; the flying-cat, puma, cat, dog, marmot, and rhinoceros, 22; the otter, 21; the Chinese monkey, raccoon, civet, hare, and rabbit, 20; the tiger and wolf, 19; the macauco, glutton, marten, fat dormouse, dromedary, giraffe, and quagga, 18; the tarsier, shrew, camel, and horse, 17; and other genera and species, without any determinate order, descending so low as to 9, 8, 7, 6, and 4. The quilled duckbill (*echidna, ornithorhynchus hystrix*) has only 12 caudal vertebræ, while the common one (*ornithorhynchus paradoxus*) has at least 20. The gibbon and vampyre bat are the only mammiferous animals, excepting the CETACEA, in which there are no coccygeal bones. It sometimes happens that a monkey or opossum loses a portion of its tail, when the truncated end is converted into a knotty excrescence, sometimes carious, always different from the taper point of the last coccygeal vertebra; and in this case it is difficult to determine the exact species.

In the CETACEA, in which the absence of *pelvis* affords no mark to distinguish the lower vertebræ into lumbar, sacral, and coccygeal, those below the dorsal may be regarded as *lumbo-coccygeal*; and their number is estimated by deducting that of the cervical and costal from the total number. The following table, which shows the number of the costal, the lumbo-coccygeal, and the total number of vertebræ, indicates that their number varies much in various *genera* of this family.

Comparative Anatomy.

	d.	Lc.	t.		d.	Lc.	t.
Lamantin	16	24	46	Porpoise	13	40	60
Dugong	18	28	46	Narwal	12	35	54
Dolphin	13	47	67	Hyperoodon	9	29	45
Tursio	13	33	58	Cachalot or White Whale	14–15	39	60
D. Globiceps	11	37	56	Greenland Whale	15	37	59
D. Griseus	12	42	61	Rorqual	14	31	52
D. Gangeticus	11	23	46				

In general, however, if we reckon the first 2, 3, or 5 vertebræ after the costal as lumbar, it may be said that the caudal vertebræ of the CETACEA vary from 22 or 25, which are the numbers respectively in the lamantin and dugong, to 34, 38, and 42, at which they may be estimated in the dolphin. We shall see that, in the dugong at least, we are guided in this estimate by the rudimental bones of the pelvis.

The coccygeal or caudal vertebræ of the MAMMALIA may be distinguished into two kinds; those which contain a canal in continuity with that of the vertebral column and sacrum, and those in which the pieces are solid. The first, which are next the sacrum, have articular, transverse, and spinous processes, distinct in proportion as the animals move their tails. The latter are generally prismatic in shape, diminish in size towards the extremity, and have only slight tubercles for muscular attachments. Animals with prehensile tails, as the American ape (*sapajous*), have above, at the base of the body of each vertebra, two small tubercles, between which pass the tendons of the flexor muscles. By means of this mechanism these animals can twine the tail round the branch of a tree with sufficient force to support the weight of the body.

The MAMMALIA with long mobile tails have often two or three small supernumerary bones placed on the lower surface of the junctions of several of the coccygeal vertebræ, from the 3d or 4th to the 7th or 8th. These sesamoid bones give attachment to muscles. In the beaver, which employs its tail as a trowel, the transverse processes are remarkable for size, while the lower spinous processes are larger than the upper ones,—an arrangement which enables it to depress the tail forcibly when it beats the ground.

The chest.

The shape of the chest in the MAMMALIA varies in general as the clavicles are present or wanting. In animals provided with clavicles, as the QUADRUMANA, bats, the squirrel, beaver, mole, ant-eater, hedgehog, and sloth, the shape of the chest approaches to the human, or is conoidal, and flattened before and behind. In those void of clavicles it is compressed laterally, from the smaller incurvation of the ribs; and the sternum makes a remarkable prominence, so that the transverse or intercostal diameter is less proportionally, and the sterno-vertebral is greater proportionally, than in man and the claviculated animals. In the long-legged animals, as the giraffe and those of the stag kind, this prominence of the sternum is sufficient to give it a keel-like appearance (*thoracæ carinatus*). In the carnivorous animals the chest presents its greatest longitudinal extent.

The sternum.

The ribs.

The number and shape of the ribs varies in the different tribes. In number, indeed, the ribs always correspond with that of the costal vertebræ. Thus, in the QUADRUMANA, ZOOPHAGA, RODENTIA, EDENTATA, and RUMINANTIA, they vary from 12 to 15 pair, with only three exceptions, the glutton, hyena, and ant-eater. In the Chinese monkey, common bat, and armadillo, they are a pair less than in

Figure 1: *
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